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Prioritizing Well Logging Based on Corrosion Risk Assessment Using an Advanced Artificial Intelligence Driven Approach

Aeshah F. Alqhtani, Mohammed A. Alameer, Nourah A. Almulla, Abdullah M. Ali, Mohamed Cassim Saloojee, Abdulrahman T. Mishkhes, and Mohammed H. Hanabi, Saudi Aramco, Dhahran, Eastern Province, Saudi Arabia

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Abstract

Downhole casing corrosion presents a persistent threat to the integrity of oil and gas wells. Effective corrosion risk management is crucial to prevent unforeseen equipment failure, compromising well performance leading to catastrophic safety events. This paper presents an innovative proactive (AI)-driven framework for predicting well-specific corrosion risk profiles to assist production engineers to identify logging interventions requirements and optimize allocated resources. The solution applies a sophisticated regression algorithm, incorporating pertinent variables encompassing well age and design, operating conditions, and environmental factors. These inputs collectively contribute to a precise characterization of each well's unique corrosion susceptibility. Following thorough data discovery, transformation, and quality control procedures, our predictive model yielded probabilistic outputs distinguishing high-risk from low-risk wells. Visualization of predicted outcomes via GIS-enabled spatial heat maps empowered engineers and decision-makers to intuitively grasp complex relationships between wells, identify clusters of elevated corrosion risk, and pinpoint candidate wells for expedited well logging evaluation. The work revealed compelling evidence supporting the effectiveness of this AI-driven strategy, demonstrating an impressive prediction accuracy rate exceeding 90% in identifying high-risk wells. The combination of established heat maps and AI-predicted well risks enabled the generation of a corrosion risk prioritization factor for non-logged wells. This allowed for focused attention on high-risk wells, guiding selective logging considerations and enhanced surveillance to ensure their integrity. Application of this methodology allowed for careful allocation of limited resources toward high-priority wells exhibiting corrosion vulnerability, effectively reducing the cost and downtime associated with unexpected failures. Conversely, lower-risk wells benefited from extended maintenance intervals, fostering substantial reductions in operational expenditures. By targeting logging interventions based on predicted corrosion risk, operators can mitigate potential asset failures, promote improved safety standards, and optimize resource allocation. This study presents a pioneering approach to corrosion risk mitigation in upstream operations by integrating machine learning, statistical modeling, in addition to geospatial analytics. Our work enhances the development of holistic digital simulations for casing corrosion, enabling optimized operations and improved field safety.

Introduction

Well integrity is a cornerstone of the oil and gas industry, defined as the application of technical, operational, and organizational solutions to reduce the risk of uncontrolled release of formation fluids throughout a well's lifecycle. A robust Well Integrity Management System (WIMS) is essential to ensure that integrity is maintained from drilling through abandonment. This comprehensive system encompasses several critical components, each playing a vital role in safeguarding well integrity. These include the selection of appropriate casing and cementing designs to achieve effective zonal isolation, the use of reliable wellhead equipment to prevent leaks, careful wellbore planning to mitigate operational risks, and the use of suitable drilling fluids to preserve formation quality. Continuous monitoring and timely maintenance are also crucial to avoid safety incidents, minimize production interruptions, and reduce operational costs. [Figure 1](#) provides an annular schematic of a typical wellbore, showing casing strings, cement layers, and wellhead components that form the basis of zonal isolation and structural protection.

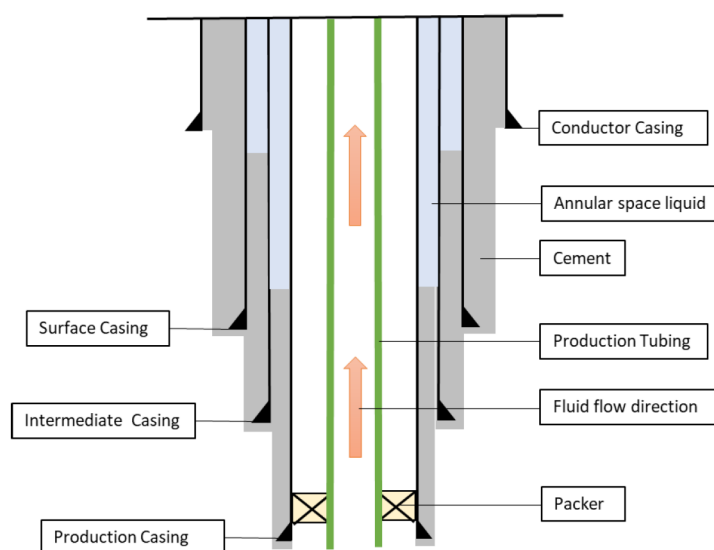


Figure 1—Annulus schematic of a typical oil well

One of the key challenges in maintaining well integrity over time is the degradation of downhole components, particularly casing corrosion, which can compromise both the structural and hydraulic integrity of the well. Integrity is maintained through a multi-barrier envelope system. As wells age, they become increasingly susceptible to such degradation, necessitating increased surveillance and more frequent assessments to ensure their continued safe operation. According to [Yakoot et al. \(2021\)](#), well integrity management in mature fields requires a state-of-the-art approach to address the unique challenges posed by aging infrastructure. Casing corrosion is not a uniform process; it can manifest in various forms, including pitting corrosion, general corrosion, and stress corrosion cracking. Pitting corrosion, for instance, occurs locally and can lead to the formation of deep pits that weaken the casing. General corrosion, on the other hand, affects larger areas and can lead to a uniform thinning of the casing wall. Stress corrosion cracking occurs under specific stress conditions and can propagate rapidly, leading to sudden failures. Understanding these different types of corrosion is essential for developing effective monitoring and mitigation strategies.

Downhole casing corrosion is typically monitored through corrosion logging, a method that employs a range of downhole tools, including ultrasonic, electromagnetic, and magnetic logging techniques, to inspect the condition of casings and tubing. These tools are essential for detecting corrosion damage, measuring metal thickness, identifying defects, and estimating the remaining life of a well. However, traditional time-based logging schedules can be inefficient and ineffective in identifying high-risk wells, especially as the number of wells requiring assessment increases. This situation creates a resource limitation dilemma,

highlighting the need for a more proactive approach to prioritize base logging based on corrosion risk predictions.

Recent advances in artificial intelligence (AI) and machine learning (ML) offer promising solutions for predicting corrosion risk in oil and gas wells. According to [Sircar et al. \(2021\)](#), the application of machine learning and artificial intelligence in the oil and gas industry has shown significant potential in improving operational efficiency and safety. By analyzing historical data and integrating various factors such as well age, design, operating conditions, and environmental factors, AI-driven models can provide accurate predictions of corrosion risk. For instance, [Al-Qahtani et al. \(2025\)](#) demonstrated the effectiveness of machine learning algorithms in identifying wells that require priority corrosion logging, achieving a high accuracy rate and highlighting the potential of AI-driven approaches in the industry. Similarly, [Macêdo et al. \(2024\)](#) explored the application of machine and deep learning techniques for predicting deformations in oil and gas pipelines, further underscoring the versatility of AI in corrosion management.

The integration of AI with traditional corrosion modeling techniques, such as corrosion rate modeling and risk-based inspection (RBI), can enhance the accuracy and reliability of corrosion prediction models. According to [Fang et al. \(2023\)](#), machine learning algorithms have been successfully applied to predict internal corrosion rates in crude oil and natural gas pipelines, offering a more precise alternative to conventional methods. Moreover, [Seghier et al. \(2022\)](#) demonstrated the effectiveness of ensemble learning techniques in predicting internal corrosion rates for oil and gas pipelines, achieving high accuracy and robust results. These studies highlight the potential of AI-driven approaches in revolutionizing corrosion prediction and management in the oil and gas industry.

In addition to improving prediction accuracy, AI can enhance the integration of geospatial analytics in corrosion risk management. According to [Madamanchi et al. \(2024\)](#), machine learning-based approaches have been successfully applied to predict corrosion levels in the oil and gas industry, providing valuable insights for risk mitigation and resource allocation. By analyzing spatial data, operators can identify clusters of wells with elevated corrosion risk and prioritize logging interventions accordingly. This spatial approach can also help in understanding the relationship between well location and corrosion risk, enabling more effective resource allocation and risk mitigation strategies.

The potential benefits of adopting an AI-driven approach to corrosion risk management are numerous. Improved safety is a primary advantage, as early detection of high-risk wells can prevent catastrophic failures and ensure the safe operation of wells. Reduced operational costs are another significant benefit, as AI-driven models can optimize resource allocation and minimize unnecessary interventions. Additionally, AI can enhance decision-making by providing operators with accurate and reliable predictions, enabling them to make informed choices about logging and maintenance activities.

The application of AI in corrosion risk management is not without its challenges, however. One of the primary challenges is the availability of high-quality data required to train machine learning models. Corrosion data can be complex and influenced by a wide range of factors, including environmental conditions, operational practices, and material properties. Ensuring the accuracy and relevance of the data is crucial for developing reliable models. Furthermore, the interpretation of model results requires expertise in both corrosion engineering and data science, which can be a barrier to implementation in some organizations.

Despite these challenges, the potential benefits of AI-driven corrosion risk management are substantial. The ability to predict corrosion risk with high accuracy allows operators to prioritize maintenance activities, reduce downtime, and extend the lifespan of wells. Additionally, the use of AI can help identify patterns and trends in corrosion data that may not be apparent through traditional analysis, leading to new insights and improved corrosion management strategies.

Problem Statement and Objective

Corrosion logging is a critical method for monitoring the condition of casings and tubing, helping to detect corrosion damage, measure metal thickness, identify defects, and estimate the remaining life of a well. However, it is not feasible to log all wells every year due to the high cost and operational constraints of downhole logging, especially in mature fields with large inventories of wells.

This challenge is further compounded by the varying levels of corrosion risk across different wells in mature fields. Production engineers must carefully prioritize which wells should undergo corrosion logging each year to ensure that well integrity is managed efficiently and that high-risk wells are not overlooked. This prioritization process is intricate, requiring a delicate balance between resource allocation and risk mitigation. Inadequate prioritization can lead to severe consequences, including increased safety risks, environmental hazards, and unexpected operational disruptions.

In this paper, we present an innovative AI-driven approach to optimize the annual selection of wells for corrosion logging. Our method integrates historical corrosion data, risk classifications derived from base logs, and geospatial features to identify wells that should be prioritized for base logging. This approach aims to enhance the efficiency and accuracy of well selection, ensuring that resources are allocated effectively and that high-risk wells receive the necessary attention. By leveraging advanced machine learning techniques and comprehensive data integration, our solution provides a robust framework for addressing the complexities of corrosion risk management in mature fields.

Methodological Framework

Data Sources and Variables

The predictive framework developed in this study was grounded on a structured dataset comprising historical data from more than 3,000 wells. These wells represent a diverse spectrum in terms of production status (active, inactive, or abandoned), geographical location, and completion characteristics. The dataset included wells spanning both onshore and offshore environments. A subset of previously abandoned wells was also utilized to support spatial analysis during risk visualization and pattern recognition.

The dataset primarily included wells that had undergone casing corrosion logging in the past, providing a baseline for model training. The final set of wells used for model training and validation was composed of over 3,000 records, although the number of wells per field varied significantly—from fewer than 30 in smaller fields to nearly 800 in larger assets.

To ensure a comprehensive understanding of corrosion risk, the feature set was designed to encompass multiple dimensions related to the well's lifecycle, completion, and subsurface conditions. The most critical features integrated into the model included:

- Well age and wellhead installation year, representing the structural aging of the well.
- Well location, which indirectly reflects regional geological and environmental exposure.
- Production-related parameters: flowing pressure, water cut, flowing rate, and pressure readings (shut-in, datum).
- Environmental indicators: presence of H_2S , total dissolved solids (TDS), and formation gradients.
- Deviation metrics, including the maximum deviation angle, which could influence flow behavior and stress points.
- Reservoir-related inputs, such as TCA (Tubing-Casing Annulus), and CCA (Casing-Casing Annulus) features.

Initial exploratory data analysis included assessment of feature distributions and their relationship to corrosion risk. Certain features were excluded during the modelling process either due to weak or non-

existent correlation with the output variable, poor data quality, or high degrees of missingness. Categorical variables or operational flags that lacked generalizability or introduced instability into the model were also filtered out. In some cases, features were dropped due to lack of availability across a sufficient number of wells, ensuring consistency in training and prediction phases.

Data missingness was another key challenge. Several wells had partial missing values across different features, in such cases, values were imputed based on temporal proximity (e.g., using the most recent available measurement), spatial similarity with offset wells, or general engineering assumptions where appropriate. These imputation strategies were carefully validated to minimize bias.

Despite certain limitations in data completeness and consistency, the overall structure of the dataset, supported by tailored preprocessing and modeling strategies, enabled the model to capture relevant patterns contributing to casing corrosion risk, serving as a practical foundation for prediction.

Data Preparation Process

The process of preparing the dataset for predictive modelling involved several critical steps to ensure data integrity, minimize noise, and enhance model generalizability as shown in Figure 2. Given the diversity in well characteristics and the historical nature of the data, thoughtful preprocessing was essential to build a reliable foundation for training. The first stage involved data cleansing, where outliers and anomalous records were reviewed and filtered. This included removing invalid entries and duplicate records.

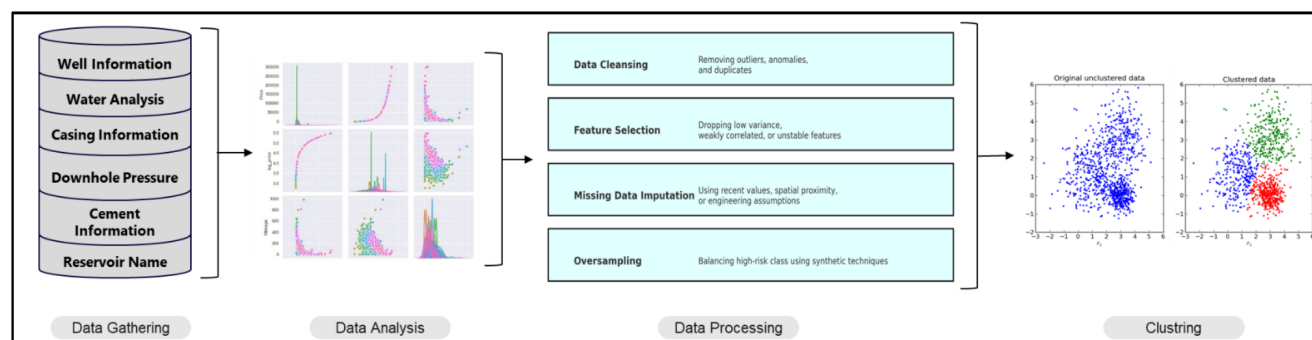


Figure 2—Data preparation process workflow.

Next, the dataset underwent feature selection and transformation. Features with minimal variance or weak correlation with the corrosion risk classification were excluded to avoid noise and reduce dimensionality. Additionally, several categorical or flag-type variables were intentionally removed due to the low interpretability, or unstable impact on model performance. Some features with known operational relevance were also excluded due to high levels of missingness.

To address missing values—common across multiple features but varying from well to well—different imputation techniques were applied. In cases where the missing feature was relatively stable over time, the most recent known value was used. For geographically clustered wells, missing values were inferred from nearby wells with similar profiles. These techniques were carefully chosen to preserve important statistical relationships without introducing artificial bias.

One key challenge in the dataset was class imbalance, with a relatively small number of wells classified as high-risk (Class 0). To address this, oversampling strategies were employed to increase the representation of the high-risk class during model training. This allowed the algorithm to better learn from rare but critical corrosion cases.

Additionally, a clustering-based segmentation approach was explored to enhance model accuracy. Wells were grouped into two main clusters based on similarities in selected input features. Each cluster was then assigned a dedicated model tailored to the unique data patterns within that group. This two-model strategy proved beneficial, especially in managing heterogeneity across wells and improving prediction robustness.

These preprocessing steps were necessary to prepare the data for modeling, given the challenges of missing values, class imbalance, and variability across wells. While not perfect, the resulting dataset was sufficient to support model training and evaluation under realistic field conditions.

Predictive Model Description

Following the data preparation phase, a range of machine learning algorithms was explored to develop a classification model capable of distinguishing wells with elevated corrosion risk. The goal of the modeling phase was to support the identification and prioritization of wells that may benefit from early intervention and targeted logging, based on predicted risk levels.

The task was framed as a supervised classification problem, with wells categorized into two main risk classes. Several candidate algorithms were evaluated, including both linear and ensemble-based methods, to identify the most suitable model for the available data and project objectives. The overall modeling workflow is illustrated in Figure 3. Model selection was based on a combination of classification performance, consistency across evaluation subsets, and interpretability.

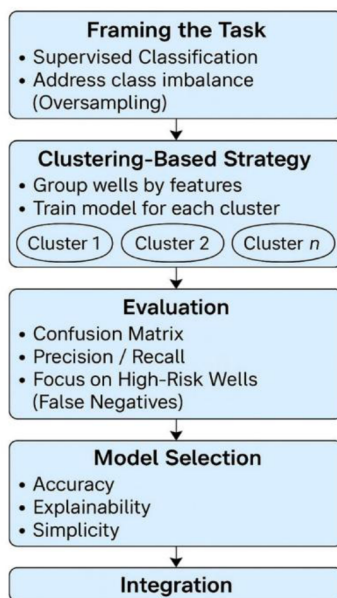


Figure 3—Overview of the model development workflow.

Due to inherent class imbalance in the dataset, techniques were applied to adjust the distribution of the minority class and reduce bias in model training. These included data-level strategies aimed at improving the model's sensitivity to wells associated with higher corrosion risk. Special consideration was given to minimizing false negatives in this category, given their potential operational consequences.

Additionally, wells were segmented into subgroups based on shared characteristics, and a tailored modeling strategy was adopted for each segment. This clustering-based approach allowed the models to better capture variations in corrosion behavior across different well types and environments, thereby improving predictive reliability.

Model performance was assessed using standard evaluation techniques common in binary classification tasks. These included confusion matrix-based metrics, as well as precision-recall analysis to better understand the trade-offs between different error types. Evaluation was conducted at the class level, and particular attention was given to performance in the high-risk class.

While multiple algorithms were considered, the final model selection favored a method that balanced predictive accuracy with simplicity and explainability. The selected model performed consistently across

test conditions and demonstrated adequate generalization to previously unseen data, forming the basis for the subsequent prioritization phase.

Results and Discussion

The predictive model developed in this study produced outputs that offered practical value for field engineers and integrity management teams. While the model was not deployed for automated decision-making, its results were integrated into a broader advisory system designed to support logging prioritization and resource planning.

In reviewing the model outputs, a number of general patterns were observed. Wells with longer operational lifespans often exhibited a higher likelihood of corrosion risk, which aligned with prior engineering assumptions. However, the model also identified several cases where relatively young wells were classified as high-risk, highlighting potential hidden vulnerabilities. This prompted further review of non-obvious cases and introduced a more nuanced understanding of how corrosion risk may develop under different conditions.

Well location also appeared to influence predicted risk levels, with certain regions showing higher concentrations of wells flagged as high-risk. These patterns were cross-referenced with historical logging data and corrosion rate observations to validate spatial correlations. It is worth noting that cementing information—widely regarded as a key factor in casing integrity—was not available in sufficient quality or coverage to be included in the model. The absence of this data likely limited the model's ability to explain certain edge cases, and its inclusion in future iterations may enhance predictive accuracy.

The outputs of the model were visualized using a geospatial dashboard that combined risk predictions with historical corrosion data to produce heat maps. Figure 4 illustrates this spatial visualization, showing the distribution of logged wells (a), historical corrosion heatmap based on those wells (b), and predicted risk class for unlogged wells (c). These visual tools allowed engineers to observe not only the risk level of individual wells, but also the spatial context of surrounding wells with known corrosion outcomes. This approach facilitated more informed decision-making, particularly when identifying base wells that had never been logged but were predicted to be both high-risk and located in historically vulnerable areas.

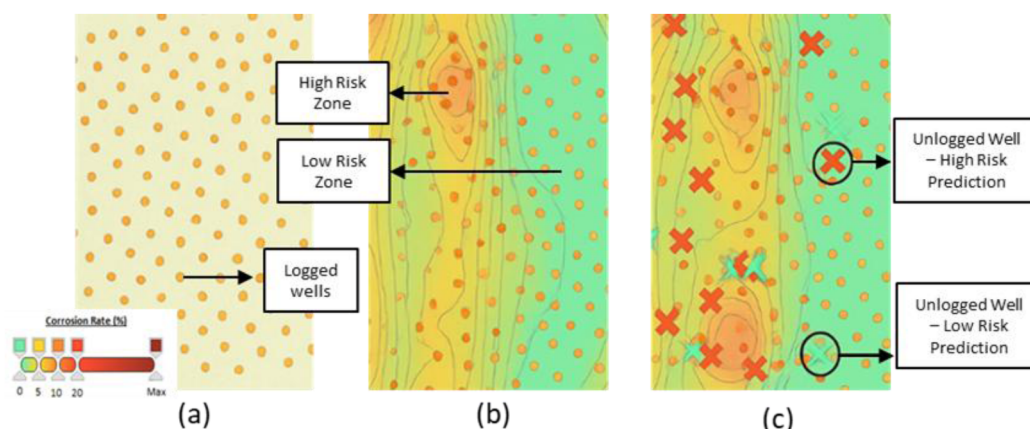


Figure 4—Geospatial visualization of corrosion risk. (a) Logged wells used as historical data references. (b) Heatmap derived from wells with confirmed corrosion outcomes. (c) Predicted risk class for unlogged wells based on the AI model output and spatial patterns.

These insights are currently being used in an advisory capacity. The tool supports prioritization by highlighting wells with higher predicted risk and spatial exposure, while still allowing engineers to apply their domain expertise and field knowledge to make final decisions on logging.

Unexpected cases were also encountered, including instances where model predictions did not align with post-logging outcomes. These cases were analyzed to identify potential gaps in the input data or modeling assumptions and contributed to an ongoing feedback loop aimed at refining the tool's logic and boundaries.

Overall, the model served as a useful addition to ongoing well integrity efforts. It supported engineers with an additional layer of insight that complemented existing practices and provided a structured way to consider corrosion risk alongside other operational factors.

Conclusion

As part of broader effort to enhance well integrity planning through data-driven tools, this study introduced a machine learning framework designed to support corrosion risk assessment and well logging prioritization. By integrating various operational and environmental features, the model offered structured insights to aid decision-making and guide engineers in identifying wells that may benefit from early intervention.

The framework demonstrated practical value by enabling more effective resource allocation and supporting planning activities through data-driven prioritization. Its use as a reference tool allowed engineers to consider predicted risk alongside field knowledge during the logging selection process.

While the current model has shown utility in operational planning, it remains under continuous development. Future improvements include incorporating new variables—when data availability and relevance are confirmed—as well as refining the algorithm itself. A structured feedback loop has been established to regularly update the model with new logging results and retrain it accordingly.

Overall, the model provided tangible support to existing workflows and opened new possibilities for enhancing corrosion monitoring strategies. Continued evaluation and enhancement are expected to strengthen its role in well integrity management.

References

- Al-Qahtani, Ahmed S., Mishkhes, Abdulrahman T., Alameer, Mohammed A., Alomair, Sammer A., and Abdulaziz A. Sultan. (2025). Machine and Systems for Identifying Wells Priority for Corrosion Log Utilizing Machine Learning Model. Paper presented at the Offshore Technology Conference, Houston, Texas, USA, May 2025. DOI: [10.4043/35533-MS](https://doi.org/10.4043/35533-MS)
- API (2015). API 581: Risk-Based Inspection Methodology. American Petroleum Institute.
- Fang, J., Cheng, X., Gai, H., Lin, S., Lou, H.: (2023). Development of machine learning algorithms for predicting internal corrosion of crude oil and natural gas pipelines. *Comput. Chem. Eng.*, **177**, 108358. DOI: [10.1016/j.compchemeng.2023.108358](https://doi.org/10.1016/j.compchemeng.2023.108358)
- Macêdo, B.S., Boratto, T.H.A., Saporetti, C.M., Goliatt, L. (2024). A Review of Deformations Prediction for Oil and Gas Pipelines Using Machine and Deep Learning. In: Bekdaş, G., Nigdeli, S.M. (eds) New Advances in Soft Computing in Civil Engineering. *Studies in Systems, Decision and Control*, vol **547**. Springer, Cham. DOI: [10.1007/978-3-031-65976-8_16](https://doi.org/10.1007/978-3-031-65976-8_16)
- Madamanchi, A., Rabbi, F., Sokolov, A.M., and Hossain, N.U.I. (2024). A Machine Learning-Based Corrosion Level Prediction in the Oil and Gas Industry. *Engineering Proceedings*, **76**(1), 38. DOI: [10.3390/engproc2024076038](https://doi.org/10.3390/engproc2024076038)
- Seghier, M.E.A.B., Höche, D., Zheludkevich, M.: (2022). Prediction of the internal corrosion rate for oil and gas pipeline: Implementation of ensemble learning techniques. *J. Nat. Gas Sci. Eng.*, **99**, 104425. DOI: [10.1016/j.jngse.2022.104425](https://doi.org/10.1016/j.jngse.2022.104425)
- Sircar, A., Yadav, K., Rayavarapu, K., Bist, N. and Oza, H. (2021). Application of machine learning and artificial intelligence in oil and gas industry. *Pet. Res.* DOI: [10.1016/j.ptlrs.2021.05.009](https://doi.org/10.1016/j.ptlrs.2021.05.009)
- Yakoot, M.S., Elgibaly, A.A., Ragab, A.M.S. et al. (2021). Well integrity management in mature fields: a state-of-the-art review on the system structure and maturity. *J Petrol Explor Prod Technol*, **11**, 1833–1853. DOI: [10.1007/s13202-021-01154-w](https://doi.org/10.1007/s13202-021-01154-w)